

S-CSIS415 BCS45 MACHINE LEARNING 2 LECTURE 1st AY 2025-2026

Module 5: Model Validation and Evaluation Metrics

ML2 Module 5: Short Quiz



Submissions

Here are your latest answers:

Question 1

What is a potential risk if a model is validated only on the same data used for training?

Choices:

- Overfitting to the training data
- Underfitting the training data
- Faster computation of results
- Better generalization to new data

Response: Overfitting to the training data

Feedback: Validating on training data risks overfitting, where the model memorizes patterns instead of learning generalizable features, leading to poor performance on new data.

Score: 1 out of 1 Yes

Question 2

Which validation method uses the same dataset for both training and test.

Choices:

- In-sample validation
- Out-of-sample validation
- K-Fold cross-validation
- Leave-One-Out cross-validation

Response: In-sample validation

Score: 1 out of 1 Yes

Question 3

What does a high F1 Score indicate about a model?

Choices:

- It balances both precision and recall effectively
- It has poor accuracy, recall and precision
- It has high recall but low precision equal to the indicate model.
- It only performs well on training and testing data

Response: It balances both precision and recall effectively

Score: 1 out of 1 Yes

Question 4

Which of the following is the correct formula for calculating the F1-Score?

Choices:

- $2 * (\text{Precision} * \text{Recall}) / (\text{Precision} + \text{Recall})$
- $(\text{Precision} + \text{Recall}) / 2$
- $\text{Precision} / \text{Recall}$
- $\text{Recall} / \text{Precision}$

Response: $2 * (\text{Precision} * \text{Recall}) / (\text{Precision} + \text{Recall})$

Score: 1 out of 1 Yes

Question 5

What is the primary purpose of model validation in machine learning?

Choices:

- To evaluate the performance of unseen data
- To train the model on the largest dataset possible
- To increase the size of the training dataset
- To reduce the number of features in the model

Response: To evaluate the performance of unseen data

Score: 1 out of 1 Yes

Question 6

The process that helps us evaluate the performance of a trained model is called:

Choices:

- Model Validation
- Model Training
- Model Deployment
- Model Optimization

Response: Model Validation

Score: 1 out of 1 Yes

Question 7

A financial institution is developing a machine learning model to predict loan defaults. The model will be used to automatically approve or reject loan applications. The dataset contains a mix of numerical and categorical features, and the loan default rate is relatively low (around 5%). The stakeholders are particularly concerned about minimizing the risk of incorrectly denying loans to creditworthy applicants.

Which of the following model validation and evaluation strategies would be MOST appropriate in this scenario?

Choices:

- Implementing stratified K-fold cross-validation and focusing on recall.
- Using accuracy as the primary evaluation metric and a simple train/test split for validation
- Employing K-fold cross-validation with balanced accuracy as the evaluation metric.
- Performing in-sample validation and optimizing the model for overall accuracy on the training data.

Response: Implementing stratified K-fold cross-validation and focusing on recall.

Feedback: stratified K-fold cross-validation ensures accuracy and representative model evaluation in the face of class imbalance, while focusing on recall aligns the evaluation with the specific business objective of minimizing the risk of denying loans to creditworthy applicants.

Score: 1 out of 1 Yes

Question 8

Why is conceptual review important in model validation?

Choices:

- It checks the logic and assumptions of the model

- It increases the dataset size in the training and testing model data
- It splits data into training and testing sets
- It removes bias from the data

Response: It checks the logic and assumptions of the model

Score: 1 out of 1 Yes

Question 9

Which evaluation metric would be most appropriate for a model predicting continuous values?

Choices:

- Regression metrics
- Precision
- Classification metrics
- Clustering metrics

Response: Regression metrics

Score: 1 out of 1 Yes

Question 10

Which metric measures the proportion of true positive predictions out of all positive predictions made by the model?

Choices:

- Precision
- Accuracy
- Recall
- F1 Score

Response: Precision

Score: 1 out of 1 Yes

Question 11

Which of the following is the correct formula for calculating Precision?

Choices:

-
 $P = TP / (TP + FP)$

-
 $P = TP / (TP + FN)$

-
 $P = TN / (TN + FP)$

-
 $P = TN / (TN + FN)$

Response: $P = TP / (TP + FP)$

Score: 1 out of 1 Yes

Question 12

A machine learning engineer is tasked with selecting the best model for predicting customer churn at a telecommunications company. They have two candidate models:

Model A: Achieves high precision (90%) but relatively low recall (60%).

Model B: Achieves moderate precision (75%) and moderate recall (75%).

The company's business objective is to minimize customer churn while also controlling the costs associated with offering incentives to customers who are predicted to churn.

Which model should the engineer select, and why?

Choices:

-
Model B, because its balanced precision and recall provide a better overall performance and reduce both false positives and false negatives.

-
Model A, because it has higher precision, which means fewer incentives will be wasted on customers who were not going to churn anyway.

-
Model A, because precision is always the most important metric in churn prediction.

-
Neither model, because both have unacceptably low recall.

Response: Model B, because its balanced precision and recall provide a better overall performance and reduce both false positives and false negatives.

Score: 1 out of 1 Yes

Question 13

What is the main purpose of model validation in machine learning?

Choices:

-
To evaluate unseen data

-
To train the model with new data

-
To visualize model outputs

-

To store training datasets

Response: To evaluate unseen data

Score: 1 out of 1 Yes

Question 14

Why is out-of-sample validation preferred over in-sample validation?

Choices:

-

It ensures realistic model performance on new data

-

It requires less data for training

-

It is faster to compute

-

It uses the same data for training and testing

Response: It ensures realistic model performance on new data

Feedback: Out-of-sample validation tests the model on new, unseen data, providing a more realistic estimate of how the model will perform in real-world scenarios, reducing overfitting risks.

Score: 1 out of 1 Yes

Question 15

A credit card company uses K-Fold Cross-Validation to test its fraud detection model on different data subsets to ensure consistent performance.

Statement: This approach helps prevent bias and gives a more reliable evaluation.

Choices:

-

True

-

False

Response: True

Feedback: True- K-Fold Cross-Validation ensures the model is tested on multiple portions of data, improving fairness and generalization

Score: 1 out of 1 Yes

Question 16

In which situation is recall more important than precision?

Choices:

-

When catching all positive cases is critical

- When false positives are very costly
- When overall accuracy is the priority
- When the dataset is balanced in precision

Response: When catching all positive cases is critical

Score: 1 out of 1 Yes

Question 17

Why can accuracy be misleading in imbalanced classification problems?

Choices:

- It ignores the performance on minority classes
- It measures only false positives
- It focuses solely on false negatives
- It always underestimates model performance

Response: It ignores the performance on minority classes

Score: 1 out of 1 Yes

Question 18

What does the F1 Score represent in classification metrics?

Choices:

- The harmonic mean of precision and recall
- The average of accuracy and recall
- The difference between precision and recall
- The sum of true positives and false positives

Response: The harmonic mean of precision and recall

Score: 1 out of 1 Yes

Question 19

Why is model validation especially critical in high-stakes domains like healthcare or autonomous vehicles?

Choices:

-

Because errors can lead to serious harm

-

Because these models require less data

-

Because these models are simpler

-

Because these models are always perfect

Response: Because errors can lead to serious harm

Score: 1 out of 1 Yes

Question 20

Which validation technique is repeated k times, each time using a different fold for testing?

Choices:

-

K-Fold Cross-Validation

-

Holdout Method

-

Train-Test Split

-

Leave-One-Out Validation

Response: K-Fold Cross-Validation

Score: 1 out of 1 Yes

Question 21

The spam detection model correctly identifies most spam emails but also labels important messages as spam. The model's recall is high, but its precision is low.

Statement: This model is performing well because it catches all spam.

Choices:

-

True

-

False

Response: False

Feedback: False-Although recall is high, low precision means too many false positives — the model incorrectly marks legitimate emails as spam.

Score: 1 out of 1 Yes

Question 22

Why does accuracy sometimes fail as a reliable metric for imbalanced data?

Choices:

-
- It can appear high accuracy even minority classes are misclassified.
-
- It always underestimates and overfitting model performance output.
-
- It focuses too much on false negatives than the false positive.
-
- It measures both recall and precision classes with is misidentified.

Response: It can appear high accuracy even minority classes are misclassified.

Score: 1 out of 1 Yes

Question 23

A data scientist trains a model using all available data and tests it on the same dataset. The model shows 100% accuracy; therefore, should we assume it's ready for deployment?

Choices:

-
- True
-
- False

Response: False

Feedback: false-Testing on the same data causes overfitting — the model memorizes training patterns instead of learning to generalize to new data.

Score: 1 out of 1 Yes

Question 24

A weather prediction model assumes temperature patterns remain constant all year. The predictions are often wrong during sudden seasonal changes.

Statement: The model failed due to poor data quality.

Choices:

-
- True
-
- False

Response: False

Feedback: False- The issue is with faulty assumptions (a conceptual error), not data quality. The model's logic didn't account for real-world fluctuations.

Score: 1 out of 1 Yes

Question 25

In a hospital, an AI model is designed to detect rare diseases. The team prioritizes recall to ensure no sick patient is missed, even if it produces some false alarms.

Statement: Prioritizing recall is appropriate for this case.

Choices:

-

True

-

False

Response: True

Feedback: True- In medical settings, missing a positive case (false negative) can be dangerous, so high recall is more critical than perfect precision.

Score: 1 out of 1 Yes

Question 26

Which validation technique divides data into k subsets, training on k-1 and testing on the fold that is left?

Choices:

-

K-Fold cross-validation

-

Holdout method

-

Leave-One-Out cross-validation

-

Stratified sampling

Response: K-Fold cross-validation

Score: 1 out of 1 Yes

Question 27

Which component of model validation focuses on removing missing values and detecting outliers?

Choices:

-

Conceptual review

-

Testing

-

Data validation

-

Model training

Response: Data validation

Score: 1 out of 1 Yes

Question 28

What is a key advantage of stratified K-Fold cross-validation?

Choices:

- Each fold has the same ratio classes as the overall dataset
- It uses only one data point for testing
- It is one piece of data is not used to train the model
- Helps improve the quality and quantity of training and testing data

Response: Each fold has the same ratio classes as the overall dataset

Score: 1 out of 1 Yes

Question 29

In which scenario is precision more important than recall?

Choices:

- Identifying spam emails
- Detecting a rare disease
- Catching all cases of fraud
- Predicting loan defaults

Response: Identifying spam emails

Score: 1 out of 1 Yes

Question 30

Which of the following is the correct formula for calculating Recall (Sensitivity)?

Choices:

- $\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$
- $\text{Recall} = \text{TP} / (\text{TP} + \text{FP})$
- $\text{Recall} = \text{TN} / (\text{TN} + \text{FN})$
- $\text{Recall} = \text{TN} / (\text{TN} + \text{FP})$

Response: $\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$

Score: 1 out of 1 Yes

